Topic 20 — World Agriculture

Covers syllabus (click to expand)

Agriculture systems | Whittlesey regions | Cereals | Fibres / sugar / oilseeds | Plantation (tea, coffee, cocoa, rubber) | Livestock | Shifting cultivation | Green Revolution (हरित क्रांति) / FAO | Von Thünen

Sources baked in: Lucent world-agriculture match-lists; NCERT Class 12 (primary activities); coaching crop-belt map; **Ghatnachakra** Agriculture & Livestock (CB–208+); UPPCS Prelims 2018–2025 **Weight:** ★★★ — plantation vs grain, coffee 2016 Brazil–Vietnam, cocoa West Africa, Kerala rubber, shifting names, citrus belt Mediterranean; teach full systems map **Last verified:** August 2026 — coaching gap-pass (Von Thünen, producer≠exporter, soybean, milk) **Current Affairs:** none mandatory (producer ranks freeze to the PYQ year when the paper quotes a year)

Current Affairs (this topic)

None mandatory. If a paper quotes a **year** (e.g. 2016 coffee), freeze **that** ranking. Do not chase every FAO revision in the teaching home.

N.0 Framework — World Agriculture

Identity: This chapter links **farming systems** to **crop belts** and the **producer ≠ exporter** trap.

First fact the **system**.

Intensive, extensive, plantation, Mediterranean, dairy, ranching and shifting are the main named systems.

Then fact the **crop climate** and the **country rank the paper froze**.

Coffee 2016-type ranks are year-frozen examples, not forever truths.

Whittlesey is the world-type agricultural map.

Von Thünen is the ideal market-distance ring model.

Do not merge Whittlesey with Von Thünen.

India crop detail lives in the Indian Agriculture chapter.

Here keep only world centres that sneak into papers, such as Kerala rubber, Assam tea and Philippines cane.

Order of this chapter

1. Systems + Whittlesey (+ Von Thünen).
2. Cereals, then fibres / sugar / oilseeds.

3. Plantation crops (tea, coffee, cocoa, rubber), then livestock, shifting cultivation, and FAO / Green Revolution.

Teach the **full systems map**. Next year can ask any Lucent crop belt, not only tea or coffee.

N.1 Agriculture (world systems)

Agriculture is the primary activity that raises crops and/or animals. Teaching first splits **subsistence versus commercial**, then **intensive versus extensive**, then the **named systems**.

Split	Meaning	picture
Subsistence	Family food; little surplus	Wet-rice monsoon Asia; shifting tropics
Commercial	Surplus for **market / export**	Prairie wheat, plantation, dairy NZ
Intensive	High labour (or capital) **per hectare**; small farms	Monsoon rice; Dutch dairy; Japanese rice
Extensive	Low labour per hectare; **huge** farms, machines	Canadian Prairie, Australian Downs, Pampas
Sedentary	Same fields year after year	Most of Eurasia
Nomadic / shifting	People or plots **move**	Sahara herders; Amazon/Congo jhum

Intensive vs extensive (सघन) (do not swap)

	Intensive subsistence	Extensive commercial grain
Land	Tiny holdings	Very large holdings
Labour	Family, high per ha	Machines, low per ha
Capital	Low–medium	High (machines, silos)
Crop	Rice (wet) or wheat–millets	Wheat, maize
Where	E / S / SE Asia	Prairies, Pampas, Downs, Ukraine–Russia steppe (स्टेपी)
Surplus	Thin; local markets	Export belt

Other named systems (one home each)

System	Climate / where	What is grown / raised	Note
Mixed farming	W Europe, US Midwest	Crops **+** livestock on the **same** farm	Fodder rotation; manure; not “only wheat”
Dairy farming	NW Europe, Great Lakes, NZ, Denmark, Netherlands	Milk, butter, cheese	Mild climate + fodder + urban (नगरीय) market; New Zealand is classic grass-export dairy

System	Climate / where	What is grown / raised	Note
Mediterranean	30–45° **west coasts** (five patches)	**Citrus, vine, olive**, winter wheat, vegetables	Winter rain + dry bright summer (2020 citrus answer **C**)
Livestock ranching	Pampas, W USA Prairie, Australian Downs, S African Veld	Cattle / sheep, **commercial**	Not nomadic herding (that is subsistence)
Nomadic herding	Sahara–Arabia–C Asia–tundra (टुंड्रा)	Camel, sheep, yak, reindeer	Moves with pasture; low density
Truck / market gardening	Urban fringes	Vegetables, fruit, flowers	High value, perishable, near city
Horticulture	Med + hill + greenhouse	Fruit, veg, flowers	Viticulture means **grape / wine**
Plantation	Humid tropics	Tea, coffee, cocoa, rubber, cane, oil palm, banana, coconut, spices	Humid-tropic estate monoculture
Shifting / slash-and-burn	Humid tropics	Millets, maize, cassava, upland rice	See shifting-cultivation card below
Collective / cooperative	Former USSR kolkhoz; Israel kibbutz; Denmark dairy co-ops	Shared marketing / land	Secondary example

Viticulture / sericulture / pisciculture / apiculture

- **Viticulture** is the cultivation of grapes, especially for wine, in France, Italy, Spain, California, Chile, Australia and the Cape.
- **Sericulture** is silk rearing; **China** leads world volume, while India's mulberry belt runs through Karnataka, Andhra Pradesh and West Bengal, with tussar, eri and muga as other silk types.
- **Pisciculture** is the rearing of fish in ponds, tanks or controlled waters.
- **Apiculture** is beekeeping for honey and related products.
- **Floriculture** is commercial flower growing; the Netherlands flower auction is the classic world hub.

Term	Recall
Viticulture	Grapes (wine) — France, Italy, Spain, California, Chile, Australia, Cape
Sericulture	Silk — **China** volume; India mulberry + tussar/eri/muga
Pisciculture	Fish rearing
Apiculture	Bees / honey
Floriculture	Flowers (Netherlands auction)

The plantation crop fact is **Tea** (not wheat, rice or maize). The world citrus belt is **Mediterranean**, not equatorial and not Kangra (कांगड़ा) / Pir (पीर) Panjal (पीर पंजाल) as the *world* answer.

Inline PYQ — UPPCS Prelims 2024, Q34

Which of the following crops is predominantly grown under plantation agriculture?

Options:

A. Wheat

B. Tea

C. Rice

D. Maize

Show answer

Ans: B — Estate crop on tropical slopes. A/C/D are intensive or extensive field grains.

Inline PYQ — UPPCS Prelims 2020, Q72

In which of the following regions of the world, the production of citrus fruits is well developed?

Options:

A. Between Kangra and Dhauladhar (धौलाधार) ranges

B. Between Pir Panjal and Himadri (हिमाद्रि) ranges

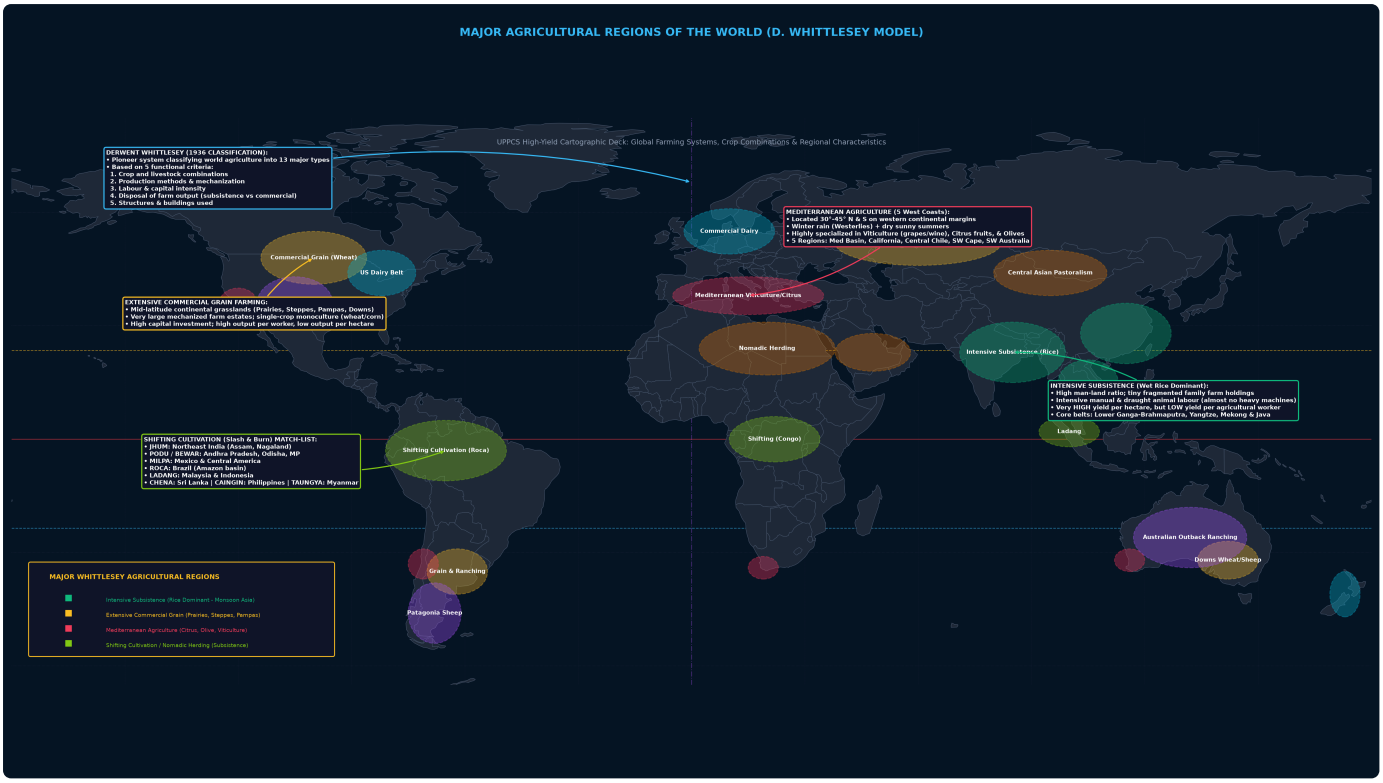
C. Mediterranean regions

D. Equatorial regions

Show answer

Ans: C — Winter rain + bright summer. Equatorial is too wet/cloudy. A/B are Himalayan distractors, not the *world* citrus belt.

N.2 Whittlesey agricultural regions



Derwent **Whittlesey (1936)** — standard 13-type world map. Learn **region + crop**, not the essay.

Type	Core belt
Nomadic herding	Sahara, Arabia, Gobi, tundra, C Asian drylands
Livestock ranching	Pampas, W USA, Australia, Veld
Shifting cultivation	Amazon, Congo, SE Asia hills, NE India, C America
Rudimentary sedentary tillage	Parts of tropics where plots are fixed but tools simple
Intensive subsistence — **rice dominant**	E / SE / S Asia deltas and coasts
Intensive subsistence — **without paddy rice**	N China, interior India wheat–millet, Nile
Commercial plantation	Caribbean, NE India–SL tea, Malaysia–Indonesia rubber/palm, Brazil coffee, W Africa cocoa
Mediterranean agriculture	Five west-coast patches (Med proper, California, C Chile, Cape, SW Australia)
Commercial grain farming	Prairies, Pampas, Downs, Ukrainian–Russian steppe
Mixed crop-and-livestock	W Europe, US Midwest Corn Belt edge
Subsistence crop-and-livestock	Parts of highlands / poorer temperate belts
Commercial dairy	NW Europe, Great Lakes, NZ

Type	Core belt
Specialised horticulture / market gardening	W Europe urban belt, California, Mediterranean coasts

Five Mediterranean west-coast patches (same climate family as citrus/vine/olive):

Patch	Where
1	Mediterranean Sea basin (Spain, Italy, Greece, Levant (लेवेंट), Maghreb)
2	California
3	Central Chile
4	Cape of South Africa
5	SW Australia (Perth side)

Von Thünen model

- Ideal isolated state: one market town, flat plain, uniform soil, farmers maximise rent.
- **Rings from the city outward (classic teaching order):**
- Intensive dairy, vegetables and flowers sit nearest the city.
- **Forestry** comes next because wood is bulky.
- Intensive field crops follow.
- Extensive **grain** comes after that.
- **Livestock ranching** sits on the outer margin.
- Distance cost decides the crop — not Whittlesey's climate-region names.
- Real world breaks the rings (climate, relief, government, refrigerated transport), but the **logic** still appears in Prelims theory stems.

N.3 World cereals

Ranks **move**. Freeze a year only when the paper quotes it. Learn **climate + belt + producer vs exporter**.

Rice

Rice needs heat plus moisture. Temperature is about **20–27°C**. Rain is about **100–200 cm**, or irrigation. **Clayey alluvial (जलोढ़)** soil that puddles is typical, with standing water. - World's main **calorie crop** for the largest number of people. - Types include **Indica** (South / South-East Asia, long grain), **Japonica** (Japan, Korea, North China — stickier), and **Javanica** (Indonesia). - The **IRRI** (International Rice Research Institute) sits at **Los Baños, Philippines**. It drove HYV rice of the Green Revolution. - **Golden rice** is genetically enriched with **Vitamin A** (2018 pair was correctly matched).

	Usual key fact
Producers (volume)	**China > India > Indonesia > Bangladesh > Vietnam**

	Usual key fact
Exporters	**Thailand, Vietnam, India, Pakistan, USA**
Trap	China is **not** the classic rice-export king — it **eats** the crop

Wheat

- Cool growing season (**~10–15°C**), bright ripening (**~20–25°C**). Rain **~50–75 cm**. Well-drained loam. Frost at ripening spoils grain.
- **Winter wheat** is sown autumn, survives mild winter (USA south, Europe, N India rabi (रबी)).
- **Spring wheat** is sown spring in **harsh-winter** belts (Canadian Prairie, Siberia).
- **Durum** is hard wheat for **pasta** (Mediterranean).

	Usual key fact
Producers (volume)	**China, India, Russia, USA, France, Canada, Australia, Ukraine**
Exporters	**Russia, USA, Canada, France, Australia, Argentina**
Trap	India/China = huge **producers**, not the Prairie-style **export** story

Commercial wheat belts

Belt	Where
Prairies	Canada + USA
Pampas	Argentina
Downs	Australia
Steppe	Ukraine / S Russia / Kazakhstan
N European plain	France (export), Germany
Indo-Gangetic rabi	India (volume, mostly domestic)

Maize (corn)

- Warm **~21–27°C**, rain **~50–100 cm**, **frost-free** growing season. Used as food, **feed**, and **ethanol**.
- The **US Corn Belt** covers Iowa, Illinois, Indiana, Ohio and neighbours — mixed farming with livestock.
- Usual volume order is **USA > China > Brazil**. Argentina is also a feed/export name.

Millets / coarse grains

- Drought (सूखा) belts of **Africa and India** grow sorghum, pearl millet and finger millet. The climate fact is **low rain**, not 200 cm tea slopes.

N.4 Fibres, sugar, oilseeds

Cotton

- Temp **~21–30°C**. Rain **~50–100 cm** (or irrigation). About **210 frost-free days**. Bright sun at picking. Black / alluvial soils.
- **Long staple** (best): **Egypt, Sudan**, USA Sea Island.
- **Medium staple:** USA Upland.
- **Short staple:India, Pakistan, China** (bulk).
- Volume names: **China, India** (often **largest area**), **USA, Brazil, Pakistan**.
- The historical **Cotton Belt** tag is the southern USA.

Jute (“golden fibre”)

- Hot humid **~25–35°C**, rain **~150–200 cm**, Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र) **alluvium**.
- World jute is almost entirely **India and Bangladesh**. Do not treat UP as a jute state (2022 trap).

Sugarcane vs sugar beet

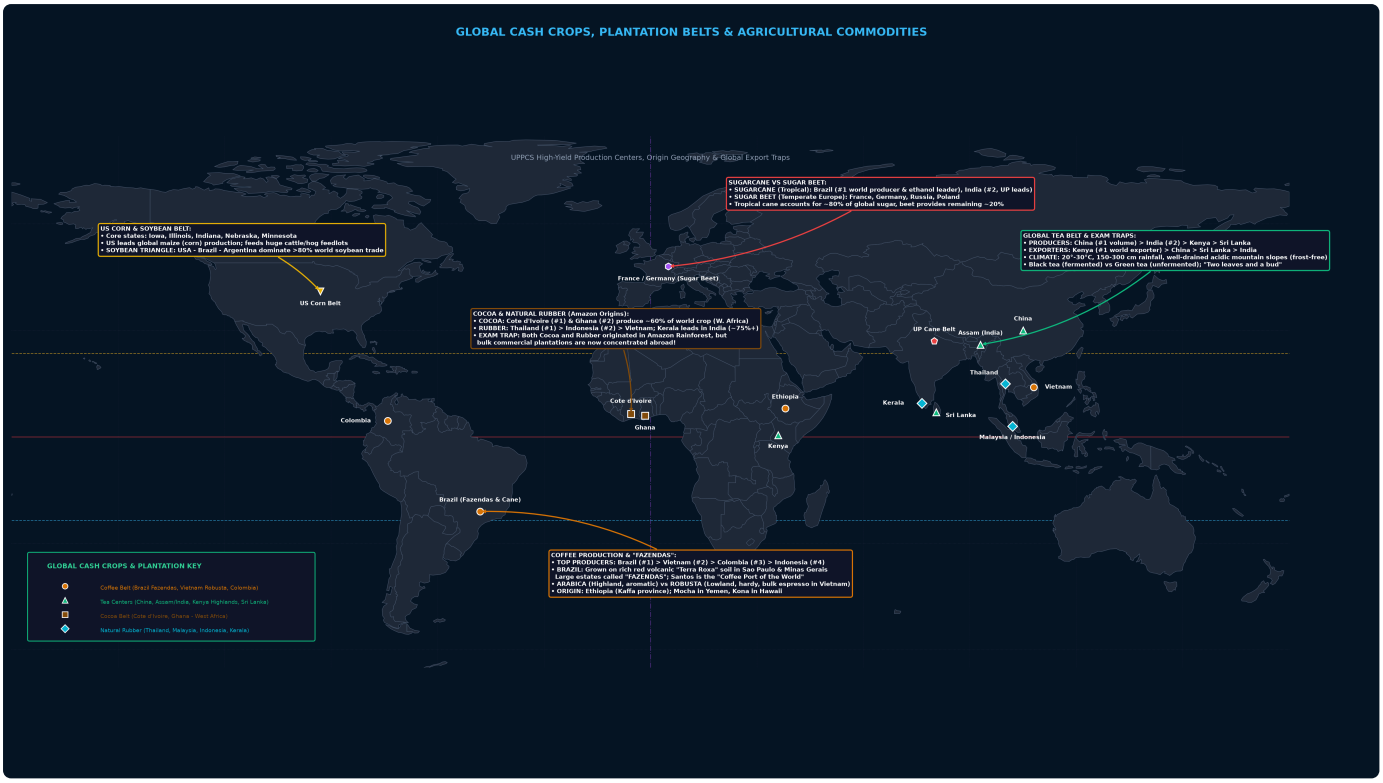
	Sugarcane	Sugar beet
Climate	Tropical; long season **10–18 months**; frost-free	Temperate; **4–7 months**
Rain / water	**~75–150 cm** or irrigation	Moderate (नरम दल) rain; Europe plains
World centre	**Brazil** (also ethanol from cane)	**France, Germany, Russia, Ukraine, USA**
Asia names	India, China, Thailand, Pakistan	Not a Kerala crop

Oilseeds and tree oils

Crop	Climate / belt	Usual leaders
Soybean	Warm temperate	**USA, Brazil, Argentina** (export triangle). China = big user
Groundnut	Tropical / subtropical	China, India, Nigeria, USA
Rapeseed / canola	Cool temperate	Canada, China, India, EU
Sunflower	Temperate	Russia, Ukraine, Argentina
Oil palm	Equatorial, high rain	**Indonesia + Malaysia** (volume). Nigeria origin story, not today’s tonnage king
Olive	Mediterranean winter-rain	**Spain, Italy, Greece**, Tunisia

> **Logic:** ** Oil palm ≠ rubber (different tree, same SE-Asia map). Olive ≠ equatorial cocoa.

N.5 Plantation crops (the system)



Feature	Fact
Climate	Tropical / subtropical; **no severe frost**
Organisation	**Estate** (large), often colonial origin
Capital + labour	High capital; **hired** labour (not only family)
Crop	**Monoculture** for **export**
Processing	Factory (फैक्टरी) on or near the estate (tea factory, rubber smoke-house, sugar mill)
Classic list	**Tea, coffee, cocoa, rubber, sugarcane, banana, oil palm, coconut, spices**
Not plantation	Wheat, rice, maize as the *predominant* system

India on the world plantation map

Crop	India	Do not write
Tea	Assam (volume), WB Darjeeling, Nilgiri (नीलगिरि)	Gujarat — Tea
Coffee	Karnataka > Kerala > Tamil Nadu (shade, Western Ghats (पश्चिमी घाट))	Assam as coffee king
Rubber	**Kerala** (Kottayam–Ernakulam)	Assam — Wheat; UP — Jute
Spices	Kerala pepper (काली मिर्च) + cardamom	
Coconut	Kerala–TN–Karnataka coast	Philippines is the *world* Spanish–US story

Crop	India	Do not write
Banana	India = huge **producer**; **Ecuador** = classic **exporter**	India = world's banana-ship king

Colonial crop stories the paper likes

Place	Crop	Credit / trap
Philippines	Coconut + sugarcane	**Spanish and Americans** (2023) — not Dutch
Indonesia / Java	Cane, coffee, rubber	**Dutch** (Hollanders)
Malaya	Rubber estates	**British**
NE India / Ceylon	Tea	British
West Africa	Cocoa	Shifted from Amazon after colonial planting
Brazil	Coffee	Portuguese / highland Terra Roxa

Inline PYQ — UPPCS Prelims 2022, Q142

Which one of the following is correctly matched?

Options:

A. Gujarat — Tea

B. Uttar Pradesh (उत्तर प्रदेश) — Jute

C. Kerala — Rubber

D. Assam — Wheat

Show answer

Ans: C — Kerala = India's rubber state. Tea = Assam/WB/Nilgiri, not Gujarat. Jute = West Bengal, not UP. Wheat = PB–HR–UP, not Assam.

Inline PYQ — UPPCS Prelims 2023, Q64

To whom does the credit go for the development of coconut and sugarcane agriculture in the Philippines?

Options:

A. French

B. Britishers

C. Hollanders

D. Spanish and Americans

Show answer

Ans: D — Spanish colony (उपनिवेश) then US period. Dutch = Indonesia; British = India/Malaya; French = Indochina.

N.6 Tea

Bush *Camellia sinensis*. Wants **20–30°C**, **150–300 cm** well-distributed rain, **well-drained slopes** (water must **not** stagnate), **acidic** soil (about pH 4.5–5.5), cheap skilled **plucking**.

- Pluck means **two leaves and a bud**.
- Tea needs hill slopes, monsoon rain and drainage. It is **not** a desert crop. It is **not** a zaid (जायद) melon.

Type	Processing	Where the puts it
Black	Fully **fermented**	India, Sri Lanka, Kenya
Green	**Not** fermented	China, Japan
Oolong	Semi-fermented	Taiwan / SE China
CTC	Crush–tear–curl (granular, tea-bag)	Indian plains/Assam bulk
Orthodox	Whole-leaf	Darjeeling, Ceylon high-grown
Region	Fact	
---	---	
China	Usual **largest producer**; original home; green tea	
India	**Assam** (Brahmaputra valley — **volume**), **Darjeeling** (GI, hill, orthodox), **Dooars/Terai (तराई)**, **Nilgiri** (नीलगिरि), Kangra, Munnar	
Sri Lanka	"**Ceylon** tea"; central highlands; high / mid / low grown	
Kenya	Leading **African** name; often among top **exporters**; highland	
Indonesia, Vietnam, Bangladesh, Turkey	Also revise	

India versus world: Assam leads Indian volume. Darjeeling leads quality / GI. Kenya is an export competitor. China is the volume king.

N.7 Coffee

Shrub *Coffea*. It wants **15–28°C**, **150–250 cm** rain, **shade**, well-drained hill soil, **no frost** and no waterlogging. The fruit is a **berry / cherry**. Processing is wet (washed) or dry.

Type	Character	Where
Arabica (*C. arabica*)	Milder aroma; higher hills (~600–1600 m)	Brazilian highlands, **Colombia**, Ethiopia, India's Ghats
Robusta (*C. canephora*)	Stronger, more caffeine, hardier, lower altitude	**Vietnam**, Uganda, lowland tropics
Liberica	Secondary example	W Africa pockets

2016 quantity (freeze this paper order): Brazil > Vietnam > Colombia > Indonesia

Country	Note
Brazil	Usual **#1**. São Paulo–Minas–Paraná highlands. **Terra Roxa** soil. Mostly **arabica**
Vietnam	**#2** in the 2016 fact. **Robusta** bulk. Central Highlands. Trap = putting it first
Colombia	Mild **arabica**; **third** in 2016 four (चातुर्याम)
Indonesia	Sumatra / Java / Sulawesi; **fourth** in 2016 four
Ethiopia	**Origin** story of arabica — **not** the 2016 tonnage king
India	Karnataka (**Kodagu, Chikmagalur, Hassan**) > Kerala (Wayanad) > Tamil Nadu (Nilgiri). Shade often **silver (रूपक) oak**

Coffee rust (*Hemileia vastatrix*) is a famous plantation disease. The Ceylon tea-shift story is linked to this rust.

Inline PYQ — UPPCS Prelims 2018, Q28

Arrange the following coffee producing countries in descending order of their coffee production (2016, quantity) and select the correct answer from the codes given below:

A. Colombia B. Vietnam C. Brazil D. Indonesia

Options:

A. D, C, B, A

B. C, B, A, D

C. B, D, C, A

D. C, A, B, D

Show answer

Ans: B — Brazil (C) → Vietnam (B) → Colombia (A) → Indonesia (D). Trap **D** puts Colombia second. Trap **C** starts with Vietnam.

N.8 Cocoa

Tree of the equatorial forest (*Theobroma cacao*). Hot, wet, **shade** of taller trees, **~150–200 cm+** rain, no long dry season. Pods grow on the **trunk** (cauliflory). Beans are **fermented** then dried.

- **Origin is the Amazon** (like rubber). **Production later shifted to West Africa**.
- Chocolate **manufacturing** (Switzerland, Belgium, Netherlands, USA) is not the same as **growing**.

Country	Fact
Côte d'Ivoire (Ivory Coast)	World's usual **#1**
Ghana (घन)	#2 belt; quality name
Cameroon	Major (2021 list)
Nigeria	West African belt
Indonesia, Ecuador, Brazil	New World / SE Asia
Latvia	**NOT** a cocoa country (2021) — Baltic **temperate forest**, not equator

Inline PYQ — UPPCS Prelims 2021, Q45

Which one of the following is NOT a major coco producer country?

Options:

A. Latvia

B. Cameroon

C. Ghana

D. Ivory Coast

Show answer

Ans: A — Paper spelling **coco** = cocoa. B, C, D are the West African belt. Latvia is the odd temperate state.

N.9 Rubber

Tree *Hevea brasiliensis* (Euphorbiaceae). Origin is the **Amazon**. It wants **25–35°C**, **>200 cm** rain evenly spread, high humidity, **no frost**, laterite (लेटराइट) or alluvial soil, and tapping labour.

- Latex comes from **tapping** the bark. The tree takes about **6–7 years** to come into tapping.

- **Henry Wickham** took seeds from the Amazon to Kew, then to Ceylon / Malaya (1870s). The production belt then **moved to South-East Asia**. Brazil is the origin, **not** today's tonnage king.
- Natural rubber is not the same as **neoprene** (synthetic — chemistry/industry fact).

Country	Fact
Thailand	Usual world leader in recent textbook years
Indonesia	Large
Vietnam	Fast growth
Malaysia	Classic **estate** region; **not** Kinta Valley
India	**Kerala** (Kottayam–Ernakulam); also TN, KA, Andamans, NE pockets
China	Growing consumer-producer
Brazil	Origin; wild/native story, not the estate king now

Kinta Valley (Malaysia) is a **tin** mining name, **not** rubber, tea or coffee. Rubber in Malaysia sits in the peninsula estate belt, not that valley name.

Inline PYQ — UPPCS Prelims 2020, Q88

In Malaysia, the 'Kinta Valley' is famous for

Options:

A. Rubber Production

B. Tea Production

C. Tin Production

D. Coffee Production

Show answer

Ans: C — Tin. A is the plantation trap because Malaysia *does* grow rubber — just not as "Kinta Valley".

N.10 Other plantation / tropical cash crops

Crop	Climate	World centre	Trap
Oil palm	Equatorial, high rain	**Indonesia + Malaysia**	Latvia / Kerala-only
Banana	Hot, wet, frost-free	India = huge **producer**; **Ecuador** (also Philippines, Costa Rica) = **export**	Equating producer with exporter

Crop	Climate	World centre	Trap
Coconut	Coastal tropics	Indonesia, Philippines, India	Philippines development = **Spanish and Americans**
Pepper (काली मिर्च)	Hot, wet vine	Vietnam / India / Indonesia (ranks move)	"Only Kerala in the world"
Cardamom	Humid hills	Guatemala + India (Kerala–Karnataka)	
Clove / nutmeg	Equatorial islands	Indonesia (Maluku / "Spice Islands")	
Tobacco	Warm, frost-careful	China, Brazil, India, USA	Not a slope-tea crop
Citrus	Med winter-rain + some subtropics	Spain, Italy, California, Brazil (orange juice)	**World belt name = Mediterranean** (2020)

N.11 Livestock, dairy, wool, silk

Ranching is **commercial**. Nomadic herding is **subsistence**. Do not put both on "Pampas".

Item	Note
Cattle numbers	**India** often #1 inventory; then Brazil, China, USA
Beef export	Brazil, Australia, USA — **not** "India exports most beef because it has most cattle" as a blind fact
Dairy	**India** leads **milk** volume (domestic). **New Zealand**, the Netherlands and Denmark lead commercial **export** dairy. Do not treat India as the cheese-ship king
Sheep / wool	**Australia** (Merino), plus China and New Zealand. Wool is not cotton
Pigs	**China**
Ranching map	Pampas cattle, Australian Downs sheep, Veld, W USA
Silk	**China** volume; India second-story (mulberry + wild silks)

N.12 Shifting cultivation

Farmers slash vegetation, then **burn** it. Ash acts as fertiliser. They crop the plot for about **1–3 years**, then leave it **fallow** so forest returns, then move to a new plot.

- It needs **low density** and **lots of land**.
- Short fallows cause **erosion and forest loss**.
- Crops include millets, maize, cassava, upland rice and vegetables — **not** estate tea.

- It is **not** plantation agriculture.
- It is **not** intensive wet-rice of the Ganga plain.

India names

Local name	Where
Jhum / jhoom	North-East India
Podu	Andhra / Odisha hills
Bewar / dahia	Madhya Pradesh
Penda	Chhattisgarh / Bastar
Kuruwa	Jharkhand
Kumari	Western Ghats / Kerala hills
Khil	Himalaya (हिमालय)
Valre / waltre	SE Rajasthan
Bringa / koman	Odisha pockets

World names (match-list gold)

Local name	Where
Ladang	**Malaysia**
Caingin	Philippines
Tamrai	Thailand
Chena	**Sri Lanka**
Humah	Java
Milpa	**Mexico / Central America**
Roca	**Brazil**
Masole	Congo basin
Ray	Vietnam
Fang	Equatorial Africa

Humid tropics: Amazon, Congo, SE Asia hills, NE India, Central America.

N.13 World Green Revolution and FAO (short)

The **Green Revolution** (हरित क्रांति) is the mid-20th-century package of high-yielding varieties, irrigation, fertiliser and plant protection that raised cereal output in many countries.

- The package logic is the same worldwide: **HYV seeds + water + fertiliser + pesticide**.
- **Norman Borlaug** led the wheat breakthrough at **CIMMYT in Mexico** and won the **Nobel Peace Prize in 1970** (not a “Nobel in Agriculture”).
- High-yield **wheat** spread from Mexico into India and Pakistan.
- High-yield **rice** centres on **IRRI at Los Baños, Philippines**, with IR8 as the classic early variety across Asia.
- The **Food and Agriculture Organization (FAO)** (एफएओ) is the UN food body with headquarters in **Rome**.
- **World Food Day** is observed on **16 October**.
- **Golden rice** is genetically enriched for **Vitamin A** — a frequent correct pair in coaching keys.

Note	Note
HYV package	Seeds + water + fertiliser + pesticide — same logic worldwide
Wheat	**Norman Borlaug**; CIMMYT **Mexico**; Nobel **Peace** (1970) — **not** “Nobel in Agriculture”
Rice	**IRRI**, **Los Baños, Philippines**
Spread	Mexico wheat into India/Pakistan; Philippines IR8 rice across Asia
FAO	Food and Agriculture Organization; HQ **Rome**; **World Food Day 16 October**
Golden rice	**Vitamin A** (correct 2018 pair)

India’s state-wise GR belt (Punjab–Haryana–western UP) is an India-agriculture fact; here remember the **world** institutions and Borlaug’s **Peace** Nobel.

N.14 GC extras — cash, livestock, terms

Producer ranks (FAO frames move by year — learn the usual leaders, not one frozen table).

Crop / item	Usual coaching leader / note
Rice + wheat volume	**China** often leads both
Rice area	**India** often has the largest harvested area
Hybrid rice	Especially associated with **China**
IRRI / Rice Gene Bank	Los Baños, **Philippines**
Sugarcane	**Brazil** leads
Sugar beet	**Russia** and **France** lead beet sugar
Ukraine sugar	Almost entirely from **beet**
Natural rubber	**Thailand** (with **Indonesia** close behind)
Coffee	**Brazil** and **Vietnam** lead; **Robusta** is an African species also grown in India

Crop / item	Usual coaching leader / note
Tea leaves	**China** leads
Tobacco	**China** leads
Silk	**China** leads
Bananas	**India** leads
Milk volume	**India** leads
Castor	**India** often leads
Coconut	**Indonesia** leads
Maize	**USA** leads
Livestock inventory	**China** often leads total headcount

Named places / crops

Note	Fact
Mocha coffee	**Yemen** (port (बंदरगाह) name)
Kona coffee	**Hawaii** (USA)
Sacramento–San Joaquin	**California** — grapes / citrus
Sugar Bowl	Classic tag for **Cuba**
Fazenda	Brazil plantation (coffee) — **not** shifting cultivation
Taungya	Agroforestry (कृषि वानिकी) / forest-farm system in **Myanmar**
Alley cropping	Crops in alleys between tree rows
Golden Crescent	Illicit opium — **Afghanistan, Iran, Pakistan** (not Iraq)
Shahtoosh	Tibetan antelope / **Chiru** hair (Tibet / Ladakh (लद्दाख) story)
Catechu (**kattha**)	From **Khair** (*Acacia catechu*) wood
Groundnut native	**Brazil** / Mato Grosso; Gambia's classic peanut economy
Alfalfa / Lucerne	Fodder legume (Fabaceae)
Heroin	From **opium**

Definitions

- **Apiculture** is beekeeping for honey and hive products.
- **Viticulture** is grape growing; **viniculture** is the wine-making side of that story.
- **Pomology** is the science and practice of fruit growing.
- **Olericulture** means vegetable cultivation — it is **not** the full label for all horticulture.
- **Horticulture** is the wider umbrella of fruits, vegetables and flowers together.

- **Green Box** support under the WTO Agreement on Agriculture covers allowed payments such as research, pest and disease control, training, and marketing or promotion — not a fertiliser or irrigation subsidy dump.

Term	Recall
Apiculture	Bees / honey
Viticulture	Grapes (viniculture = wine side)
Pomology	Fruits
Olericulture	Vegetables only
Horticulture	Fruits + vegetables + flowers
Green Box (AoA)	Allowed support — research, pest control, training, marketing

Livestock / dairy: India is the milk **volume** leader. Mad-cow disease damaged beef's "safe food" image in Europe and America. Commercial viticulture is classic **Mediterranean** (about 85% of grapes go to wine in that story).

Confused pairs (cash / terms). The **Golden Crescent** is Afghanistan–Iran–Pakistan, not Iraq. A Brazilian **fazenda** is a plantation, not jhum. **Mocha** coffee is Yemen's port name, not Brazil's lead story. **Shahtoosh** is chiru hair, not merino wool. **Olericulture** means vegetables, not fruits.

Plantation note: Ministry lists differ (Agri: coconut/cocoa/arecanut/cashew vs Commerce: tea/coffee/rubber/spices) — Prelims often treats tea/coffee/rubber/spices/cane as plantation family.
